

Fig3.14 *Mutual Induction*

LENZ'S LAW

A change in current produces an emf, whose direction is in such a way that it opposes the change in current.

Fleming's Right Hand Rule

The direction of induced EMF in generators and alternators (Dynamically induced emf) is known by Fleming's Right hand rule.

Stretch the thumb, forefinger and middle finger mutually at right angles [90°] to each other.

If the thumb indicates the direction of motion of the conductor, the forefinger indicates direction of the magnetic flux, then the middle finger indicates the direction of the induced EMF.

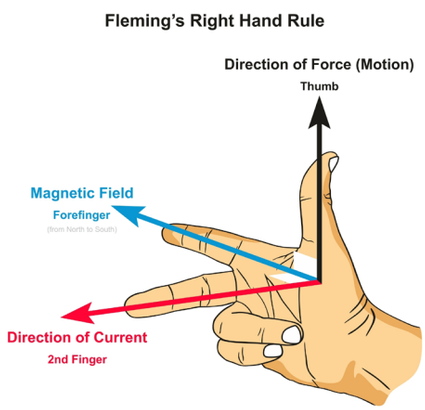


Fig 3.15 *Right Hand Rule*

3.6 HYSTERESIS LOOP

Take a piece of iron bar AB and magnetise the same by placing it within the field of solenoid. The field H produced by the solenoid is called the magnetising field. The field (H) can be increased (or) decreased by increasing (or) decreasing the current through it. Let 'H' be increased slowly from zero to a maximum value and the corresponding value of flux density (B) be noted. If we plot the relation between H and B, OA is obtained. The material becomes magnetically saturated at point A and has the maximum flux density induced in it (H = OM).

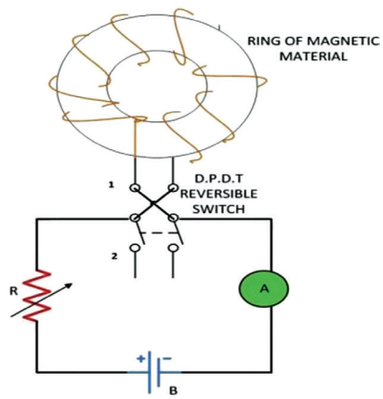


Fig 3.16 *Hysteresis Loop circuit*

Now if 'H' is decreased slowly by decreasing the current in the solenoid, the flux density(B) will not decrease along AO but will decrease less rapidly along AR_1. When H is made to be zero, at that time, B will not be zero but will have the value OR_1. It means that on removing the magnetising force, H the iron bar is not completely demagnetized. This value ($B = OR_1$) is the retentivity of the material (Residual magnetism).

To demagnetise the bar, we have to supply the force H in the opposite direction.

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